Load-bearing claim audit

This audit checks source scope, not clinical truth in the child. The full ledger is `data/evidence.csv`. Quotes below are short retrieved passages; proposed causal extensions remain visible in `data/edges.csv`.

Public Asn1002Lys record is uncertain, not verified patient pathogenic genotype

Verdict. Supported for retrieved record

Supporting passage / location. ClinVar VCV004600147: “Uncertain significance”; title `NM_001211.6(BUB1B):c.3006T>A (p.Asn1002Lys)` [E02]

Consequence. Preserve DNA-identity/phase uncertainty

Human K668 is a real residue in the retained segment

Verdict. Supported by reference and primary mechanism

Supporting passage / location. UniProt reference sequence at 668 is lysine; SIRT2 paper: “Deacetylation of lysine 668 (K668) of BubR1 inhibits ubiquitylation” [E04,E09]

Consequence. Site exists, but exact-mutant rescue remains untested

SIRT2/NMN evidence proves drug rescue of case alleles

Verdict. Unsupported extension

Supporting passage / location. SIRT2 paper reports stability and model-specific abundance/survival experiments, not either supplied allele [E09,E10]

Consequence. No such claim in report; full chain has gaps

Localized B56 recruitment can rescue a model alignment defect

Verdict. Supported in cited models

Supporting passage / location. “artificially targeted PP2A-B56s to the kinetochores was able to rescue chromosome misalignment in MVA patient cell lines” [E12, discussion]

Consequence. Retain strongest bypass experiment, not a medication

Alignment rescue proves faithful divisions

Verdict. Unsupported extension

Supporting passage / location. Source uses congression/fiber assays, including MG132 context [E12,E13]

Consequence. Require live completed divisions and selection controls

Rapamycin is proven to restore tissue in the 2026 fly model

Verdict. Unsupported extension

Supporting passage / location. “A similar result was obtained by feeding larvae with the TOR inhibitor Rapamycin”; surrounding endpoint is neuroblast number [E43 p12]

Consequence. Cell preservation only

Related genetic TOR/autophagy manipulations failed distal tissue endpoints

Verdict. Supported in cited model

Supporting passage / location. “none of the genetic interventions aimed at increasing autophagy was able to rescue the number of Pros+ cells or the size of the larval brain” [E43 pp12-13]

Consequence. Do not equate survival and tissue benefit

Arimoclomol is a universal mutant-protein chaperone rescue

Verdict. Contradicted as generalized assertion

Supporting passage / location. “we did not see correction of cholesterol storage by arimoclomol ... at doses up to 0.5 mM” [E48 published PMC abstract/results]

Consequence. Context-dependent, not proven for BUBR1

Arimoclomol's approved clinical effect establishes a precise HSP/BUBR1 mechanism

Verdict. Unsupported extension

Supporting passage / location. Label: “The mechanism(s) by which arimoclomol exerts its clinical effects in patients with NPC is unknown.” [E51 section12.1]

Consequence. Keep experimental HSP anchor distinct from clinical MoA

Potassium citrate is evidence-based for all congenital nephrocalcinosis

Verdict. Unsupported extension

Supporting passage / location. Label: “The pivotal Urocit-K trials were non-randomized and non-placebo controlled”; “Safety and effectiveness in children have not been established.” [E35 sections14,8.4]

Consequence. Conditional chemistry/formulation branch only

Larotrectinib targets TRK

Verdict. Supported

Supporting passage / location. “Larotrectinib is an inhibitor of the tropomyosin receptor kinases (TRK), TRKA, TRKB, and TRKC.” [E40 p15]

Consequence. Still requires an actual oncogenic tumor fusion/dependency

Metformin uniformly raises BUBR1

Verdict. Not supported as uniform

Supporting passage / location. Same study reports BUBR1/MAD2 *deregulation* with glucose-dependent genotoxic stress; the effect is cell-type dependent and bidirectional (raised in some, lowered in others) [E23]

Consequence. Frame metformin as bidirectional, context-dependent; direction is the weakest link

Arimoclomol activates HSP90 to refold BUBR1

Verdict. Unsupported extension

Supporting passage / location. Arimoclomol is a heat-shock-response co-inducer (HSF1 → HSP70/HSP90), not an HSP90 inhibitor; no BUBR1 bridge, and one NPC study showed no HSP70 induction [E47, E48]

Consequence. Keep as a stability hypothesis, not a demonstrated refolding mechanism

Hydroxychloroquine increases functional BUBR1

Verdict. Supported only as senescence-specific level increase

Supporting passage / location. HCQ raised BubR1 only in late-passage senescent cells by blocking lysosomal degradation; everolimus lowered it [E54]

Consequence. Degradation-blocking, not synthesis or proven function; neuromuscular caution

Amikacin is a safer aminoglycoside for read-through

Verdict. Partly supported / cautioned

Supporting passage / location. CPIC: avoid aminoglycosides in MT-RNR1 m.1555A>G/m.1494C>T/m.1095T>C carriers; no intra-class safety ranking supported [E53]

Consequence. Screen MT-RNR1 first; read-through for p.Leu737* unconfirmed

Olaparib is indicated by BUB1B-driven instability

Verdict. Unsupported extension

Supporting passage / location. CIN is not homologous-recombination deficiency; olaparib is HRR/BRCA-selected with no RMS/BUB1B indication [E24, E55]

Consequence. Requires a demonstrated tumor repair dependency

Gentamicin reporter concentration was 800 mg/mL

Verdict. Contradicted transcription artifact

Supporting passage / location. Original journal HTML uses 800 micrograms/mL; PDF extraction corrupted the micro symbol [E18 methods]

Consequence. Correct units retained; no child-dose conversion

No sufficiently supported constitutional approved candidate identified

Verdict. Coverage-limited synthesis

Supporting passage / location. No retrieved path joins exact alleles, functional rescue, feasible exposure and tumor-safe tissue benefit

Consequence. Not a claim of exhaustive absence or evidence of no effect

The most consequential limitations are the exact-allele functional gap, absence of a normal-versus-tumor window, and tissue/exposure transfers. Additional citations to the same first mechanistic step would not close them.